RUNGTA INTERNATIONAL SKILLS UNIVERSITY

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SCHOOL OF INFORMATION TECHNOLOGY

BACHELOR OF COMPUTER APPLICATION

<table>
<tr><td>PS No.</td><td>8</td></tr>
<tr><td>Project Title</td><td>Password Strength Checker</td></tr>
<tr><td>Problem Statement</td><td>Build a program that checks a password against security rules and classifies it as Weak, Medium or Strong, giving suggestions to improve it.
Language: Python 3 Concepts: Strings, Sets, Conditions, Regex
Note: One point per rule: length >= 8, uppercase, lowercase, digit, special character (@#$%&*!). Score 0-2 Weak, 3-4 Medium, 5 Strong.</td></tr>
<tr><td>Test Cases / Case Statements (4 cases)</td><td>
<table>
<tr><th>INPUT</th><th>EXPECTED OUTPUT</th></tr>
<tr><td colspan="2">Test Case 1: Normal Input</td></tr>
<tr><td>Password: abc123</td><td>Strength: Weak (2/5)
Suggestions: use at least 8 characters, add an uppercase letter, add a special character</td></tr>
<tr><td colspan="2">Test Case 2: Boundary / Minimum Input</td></tr>
<tr><td>Password: Hello123</td><td>Strength: Medium (4/5)
Suggestion: add a special character</td></tr>
<tr><td colspan="2">Test Case 3: Invalid Input</td></tr>
<tr><td>(a) Password: (empty)
(b) Password: Hello 123</td><td>(a) Password cannot be empty.
(b) Password must not contain spaces.</td></tr>
<tr><td colspan="2">Test Case 4: Different Valid Scenario</td></tr>
<tr><td>Password: Hello@123Xy</td><td>Strength: Strong (5/5)
No suggestions. Your password is secure.</td></tr>
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</td></tr>
<tr><td>Bare Minimum Requirements</td><td>1. Write the solution in Python 3 (one .py file) and run it from the console.
2. Check length, uppercase, lowercase, digit and special character.
3. Use the re module for at least one pattern check.
4. Classify the password as Weak, Medium or Strong using conditions.
5. Show which rules failed, with suggestions.</td></tr>
<tr><td>Instruction</td><td>1. Understand the problem statement and define the problem clearly.
2. Develop the solution using proper logic and programming concepts.
3. Implement all required features and pass the given test cases.
4. Test and debug the project to ensure correct output.
5. Prepare the documentation/report and upload the code and report to GitHub.
6. Complete and submit the project within the given deadline.</td></tr>
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